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Chapter: 1899 D

1. Initial Setup & Definitions

music notes 1 = $2^k = a$

music notes 2 = $2^l = b$

music = a^b

i.e. $(2^k)^{2^l} = 2^{k \times 2^l}$

there are n music notes stored in arr

so note i is $n_i = 2^{arr_i}$

2. Condition for Valid Pairs

now we need to find how many pairs exist such that the combo of

$$(n_i, n_j) == (n_j, n_i)$$

i.e.

$$n_i^{n_j} == n_j^{n_i}$$

3. Mathematical Derivation

$$(2^{arr[i]})^{2^{arr[j]}} = (2^{arr[j]})^{2^{arr[i]}}$$
$$2^{arr[i] \times 2^{arr[j]}} = 2^{arr[j] \times 2^{arr[i]}}$$

Since base of 2 is some:

$$arr[i] \times 2^{arr[j]} = arr[j] \times 2^{arr[i]}$$

Let $arr[i] = x$ and $arr[j] = y$:

$$x \times 2^y = y \times 2^x$$

4. Case Analysis

If $x == y$ then every is valid so:

$$freq(x)(freq(x) - 1) / 2$$

Case 2: If $x < y$ and $x \neq y$

$$y \times 2^x = x \times 2^y$$

$$y / x = 2^{y-x}$$

$$y = x \times 2^{y-x}$$

Let $y - x = d$

$$y = x + d$$

$$x + d = x \times 2^d$$

$$d = x \times 2^d - x$$

$$d = x(2^d - 1)$$

$$x = d / (2^d - 1)$$

$2^d - 1 > d$ for all d except 1 for which $x = d / (2^d - 1)$

for $d = 1$, $x = 1$, $y = 2$ so $\{1, 2\}$ is valid

5. C++ Code Implementation

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(nullptr);

    int t;
    cin >> t;

    while(t--) {
        int n;
        cin >> n;

        vector<long long> arr(n);
        map<long long, long long> mp;

        for(int i = 0; i < n; i++) {
            cin >> arr[i];
            mp[arr[i]]++;
        }

        long long ans = 0;
        for(auto &it : mp) {
            long long x = it.second;
            ans += x * (x - 1) / 2;
        }

        ans += mp[1] * mp[2];
        cout << ans << "\n";
    }
    return 0;
}
```

